

# Anthony M. Alam

Ottawa, Ontario, Canada | Open to relocating within Canada and to the U.S.

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## EDUCATION

### University of Ottawa

Bachelor of Applied Science in Software Engineering - GPA: 3.96/4.00

Ottawa, Ontario

Graduation: December 2027

## EXPERIENCE

### Cisco Systems

Software Engineer Intern (Telemetry and Programmability)

Ottawa, Ontario

May 2026 - Present

- Designing agentic AI workflows to automate observability tasks across distributed routing systems, built on model-driven programmability foundations (YANG, gNXI, NETCONF, RESTCONF).

Software Engineer Intern (IOS XE)

January 2025 - April 2025

- Led the development of a Python (Flask) code injection tool adopted by 75+ engineers, allowing previously unreachable conditions to be tested and increasing file line and branch coverage by 22% and 13% on average.
- Architected and re-implemented an optimal ConditionBlock graph data structure to support DAG-based flow analysis of C code, improving run time by 18% and cutting memory usage by 20%.
- Developed a library of 15 reusable APIs and algorithms for ConditionBlock graphs, already leveraged across 20+ Python files to support graph traversals, conditional path generation, and SQLite integration.
- Designed and implemented automated pre-commit routines with Python (pyATS) and YAML, covering all known edge cases and eliminating a 2-hour manual upgrade/downgrade process for Cisco Catalyst high availability switches.

Software Engineer Intern (IOS XR)

May 2024 - August 2024

- Led the development of an automated Python (pyATS) test suite for gRPC/gNMI requests, ensuring contract compliance in a multimillion-dollar customer deployment.
- Increased code coverage (+5% line, +4% branch) by developing unit tests in C (Cmocka), helping the team achieve its 80% coverage goal.
- Upgraded defect detection processes and the long-term maintainability of 100+ C files by integrating ASan and Valgrind analyses into nightly sanity runs.

### Ford Motor Company

Software Developer Intern

Ottawa, Ontario

September 2025 - December 2025

- Spearheaded the development of a C++ SOA proxy service feature enabling hotspot credential read/write operations across 400,000+ next-generation Ford and Lincoln vehicles.
- Developed a custom parser built with regular expressions to identify Wi-Fi service failures across millions of lines of ADB device logs, enhancing efficiency by 24% compared to the previous implementation.

## PROJECTS

### Grant-Winning Accessibility Tool (for Rutgers University) | JavaScript, HTML, CSS, OpenAI API

[GitHub](#)

- Spearheaded the development of a full-stack chatbot that evaluates the accessibility of Rutgers University online courses, earning a grant for AI innovation from the University of Ottawa.
- Built a custom CSV parsing algorithm to process thousands of lines of course data, enabling faculty to upload course content and instantly receive tailored accessibility recommendations based on the WCAG 2.1 guidelines.

### PlayItAgain: Catan Game Save iOS App | Swift, SwiftUI, Core Image, YOLOv8, Core ML

In Development

- Building an iOS app that digitizes physical Catan boards, cards, and game state from a photo, allowing players to save and resume games across sessions.
- Designing a Core Image perspective-correction pipeline and an on-device YOLOv8 + Core ML detector to recognize terrains, number tokens, and player pieces.

## TECHNICAL SKILLS

**Programming Languages:** Python, C/C++, Java, JavaScript, SQL, Bash, YAML, HTML, CSS

**Frameworks & Tools:** Flask, React, PostgreSQL, NoSQL, pyATS, JUnit, Cmocka, Git, GitHub, Docker, CI/CD, REST API, Unix/Linux (RHEL), VS Code, IntelliJ, Eclipse, Android Studio, Copilot, Claude Code, Jenkins, AWS

**Engineering Skills:** Full-Stack Development, Automation Testing, OOP, Networking (HTTP, IP, TCP, UDP), SDLC, Distributed Systems, Cloud Computing, Debugging (GDB), Code Review, Agile Development, Complexity Analysis

**Relevant Coursework:** Data Structures & Algorithms, Software Engineering Design Patterns, Operating Systems